

# VENKATESH BHUKYA

Cohoes, NY (Remote) | +1 (518) 308-8759 | [venkatesh.bhukya@applywizard.ai](mailto:venkatesh.bhukya@applywizard.ai) | [Portfolio](#)

## PROFESSIONAL SUMMARY

- Data Engineer with 5+ years of experience developing cloud data pipelines and platforms across banking, healthcare, and retail data environments, supporting ETL/ELT, Big Data processing, analytics, and Business Intelligence workloads.
- Builds batch and streaming data solutions using Python, SQL, PySpark, Apache Spark, Kafka, Airflow, and dbt, with hands-on experience across AWS, Azure, GCP, Snowflake, Databricks, BigQuery, Redshift, and Synapse Analytics.
- Strengthens data operations through dimensional modeling, incremental processing, query optimization, data quality, reconciliation, governance, and DataOps, reliably processing 1M+ daily records for analytics and reporting.
- Connects Data Engineering with business needs, collaborating with database, risk, healthcare, retail, compliance, and Business Intelligence teams to deliver governed datasets for financial reporting, healthcare analytics, retail insights, and decisions.

## TECHNICAL SKILLS

**Programming & Database Engineering:** Python, SQL, PySpark, Scala, Pandas, Bash, PostgreSQL, SQL Server, Stored Procedures, Query Optimization, Performance Tuning, REST APIs

**Data Engineering & Integration:** ETL/ELT, Data Pipelines, Data Integration, Data Transformation, Batch & Streaming Processing, Incremental Loads, Change Data Capture (CDC), Source-to-Target Mapping, Schema Evolution, Data Migration

**Big Data & Distributed Processing:** Apache Spark, PySpark, Apache Kafka, Hadoop, HDFS, Hive, Distributed Processing, Partitioning, Parallel Processing

**Cloud & Data Platforms:** AWS (S3, Glue, Lambda, EMR, Redshift, Kinesis), Azure (Data Factory, ADLS Gen2, Databricks, Synapse Analytics, Azure SQL), GCP (BigQuery, GCS, Dataflow, Pub/Sub, Dataproc, Cloud Composer), Snowflake

**Data Warehousing & Modeling:** Data Warehousing, Data Lakes, Lakehouse Architecture, Dimensional Modeling, Star Schema, Snowflake Schema, Fact & Dimension Modeling, Data Marts, OLAP

**Orchestration & DataOps:** Apache Airflow, dbt, Git, GitHub, Jenkins, Azure DevOps, CI/CD, Docker, Kubernetes, Terraform, Pipeline Monitoring, Release Automation

**Data Quality & Governance:** Data Quality, Data Validation, Schema Validation, Data Reconciliation, Data Lineage, Metadata Management, Data Governance, Data Catalog, Data Observability

**Security & Compliance:** IAM, RBAC, Data Encryption, Access Controls, Data Privacy, HIPAA, GDPR, CCPA

**Analytics & Business Intelligence:** Power BI, Tableau, Business Intelligence, Data Analytics, Dashboard Development, Reporting

## PROFESSIONAL EXPERIENCE

### Data Engineer

Mar 2025 - Present

Truist Bank | USA | Remote

- Engineered reusable ETL/ELT pipelines in Python and PySpark to consolidate transaction, customer, and account data into AWS, cutting downstream data preparation time by 28% for risk and reporting teams.
- Optimized SQL transformations and Redshift data models behind high-volume financial reporting, reducing average query runtime from 11 to 7 minutes while preserving reconciliation accuracy.
- Automated incremental ingestion for 10+ upstream banking feeds through AWS Glue and Airflow, replacing repetitive full loads and shortening the overnight processing window by 24%.
- Strengthened production reliability by adding validation, dependency, retry, and monitoring controls to critical data pipelines, improving successful scheduled executions from 94% to 98%.
- Streamlined Spark processing on EMR by revising partitioning and transformation logic for transaction datasets, lowering recurring compute consumption by approximately 18% without affecting delivery SLAs.
- Established source-to-target reconciliation and schema checks with Python and SQL for customer and account datasets, reducing recurring data-quality exceptions by 21% before warehouse consumption.
- Secured sensitive financial datasets through IAM-based access controls and encryption across S3 and warehouse layers, closing 100% of identified access-control gaps during internal data reviews.
- Partnered with risk, finance, analytics, and Business Intelligence teams to redesign frequently used Snowflake/Redshift datasets, reducing manual report preparation by roughly 6 hours per reporting cycle.

### Data Engineer

Mar 2024 - Feb 2025

Molina Healthcare | USA | Remote

- Developed Azure Data Factory pipelines that unified claims, member, provider, and eligibility feeds in ADLS Gen2, reducing data-delivery delays by 26% for healthcare analytics and reporting teams.
- Transformed approximately 1M+ daily healthcare records in Databricks with PySpark, standardizing inconsistent source fields and lowering downstream data-quality exceptions by 22%.
- Designed dimensional models in Synapse Analytics for claims and member reporting, simplifying complex SQL access patterns and improving commonly used analytical query performance by 27%.
- Orchestrated 15+ recurring ETL workflows through Azure Data Factory and Airflow, introducing dependency and recovery controls that increased on-time pipeline completion to 96%.
- Reduced processing overhead by replacing full dataset refreshes with incremental loading for high-volume eligibility and claims feeds, shortening recurring batch execution by 23%.
- Implemented reconciliation and schema-validation rules between source systems and curated datasets, decreasing claims-related data discrepancies by 14% before Business Intelligence consumption.
- Protected healthcare information with role-based access, encryption, and governed data-layer controls, maintaining 99% compliance with required pipeline access reviews for assigned datasets.
- Collaborated with claims operations, compliance, analytics, and Power BI teams to translate reporting requirements into curated datasets, cutting recurring manual data preparation by 5+ hours per reporting cycle.

## Data Engineer

Jan 2020 - Apr 2022

Augur Talentcare Pvt Ltd | India | Remote

Client Domain: Retail Data / Analytics

### Project: Retail Data Integration & Analytics Platform

- Built Python and SQL ETL pipelines that consolidated sales, product, inventory, and order data into BigQuery, reducing manual data preparation by 25% for retail analytics and reporting teams.
- Migrated approximately 700 GB of historical sales and product data from relational and file-based sources into GCP, validating source-to-target totals and improving access to historical retail datasets.
- Processed daily order and inventory datasets with PySpark and Dataflow, restructuring transformations and partitioning logic to reduce recurring batch-processing time by 20%.
- Scheduled 12+ dependent data workflows through Airflow and Cloud Composer, adding failure handling and restart controls that improved successful scheduled completion to 97%.
- Integrated event-driven sales and inventory feeds through Kafka and Pub/Sub, reducing data-availability delays by approximately 30% for operational analytics.
- Modeled sales, product, store, and inventory data using dimensional structures in BigQuery, improving frequently executed Business Intelligence query performance by 10%.
- Resolved source-to-target inconsistencies through SQL reconciliation and schema validation, lowering production data exceptions by 16% across critical retail datasets.
- Coordinated with data analysts, QA, Business Intelligence, and retail stakeholders to address mapping and ETL defects before release, reducing post-release data issues by 12%.

---

## PROJECTS

### Banking Data Audit & Exception Management

- Devised an audit-trail process capturing source counts, load status, and reconciliation results for financial datasets, reducing manual audit preparation by 9% across scheduled reporting cycles.
- Created SQL-driven exception views that grouped unmatched transaction records by failure reason, helping operations teams resolve recurring data issues 19% faster without searching raw pipeline logs.
- Enabled historical reprocessing of corrected financial records through parameterized Python routines, recovering affected reporting data within 2 hours instead of requiring complete dataset reloads.

### Healthcare Reference Data Standardization

- Consolidated inconsistent provider and plan reference values into governed lookup structures using PySpark and SQL, reducing downstream mapping exceptions by 17% across healthcare reporting datasets.
- Applied effective-date handling for changing member and provider reference attributes, improving historical reporting consistency across 4 analytical subject areas without overwriting prior values.
- Published reusable reference datasets through Synapse for analytics and Business Intelligence consumers, eliminating approximately 15% of duplicate transformation logic maintained across downstream reporting workflows.

### Retail Data Configuration & Reusable Ingestion Framework

- Formulated configuration-driven ingestion logic in Python to support 4 recurring retail source patterns, reducing source-specific code changes when new sales and inventory datasets were introduced.
- Normalized product, store, and order field definitions through reusable SQL mapping templates, decreasing transformation-related defects by 11% during QA and downstream reporting validation.
- Documented dataset dependencies, transformation rules, and operational handoffs for the retail ingestion framework, shortening troubleshooting time by approximately 13% during support activities.

---

## EDUCATION

Master of Science in Computer Science

Dec 2023

Eastern Illinois University | Charleston, IL

---

## CERTIFICATIONS

- Google Cloud Certified: Associate Cloud Engineer - **Google Cloud**
- AWS Certified Data Engineer - Associate - **AWS**
- Microsoft Certified: Azure Data Fundamentals (DP-900) - **Microsoft**
- IBM Data Engineering Professional Certificate - **Coursera**
- Data Engineering, Big Data, and Machine Learning on GCP Specialization - **Coursera**
- Data Warehousing for Business Intelligence Specialization - **Coursera**